



Impact of anthropogenic activities on the microplastic pollution status of an urbanized wetland

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Abstract Due to their environmental ubiquity, persistence, and potential effects on living organisms, microplastic pollution has become a global concern. Urban wetlands, catchment basins within cities, are one of the final sinks of these pollutants. We studied the occurrence, abundance, distribution, and physical and chemical characteristics of MPs to determine the impact of anthropogenic activities on the microplastic pollution status in an urbanized wetland. Surface water samples from seven sites inside the urbanized wetland Laguna del Camarón were collected during two climatic seasons (rainy and dry), followed by organic-matter digestion, filtration, microscopic counting and classification by shape and color. Subsequently, the polymers were identified with

Fourier-Transform Infrared (FTIR) spectroscopy. MPs concentration was significantly higher in the dry season (2.05 particles/L) than in rainy season (1.17 particles/L), with the higher values occurring in the small inner lagoon (LC6 and LC7), which, according to the Pollution Load Index (PLI), has been considered as unpolluted to moderately polluted and highly polluted with MPs. Fiber shape and black and blue colors are predominant, with polyester (PES), polyethylene (PE), polypropylene (PP), and polystyrene (PS) polymers mainly occurring in this ecosystem consistent with nearby recreational and construction activities. This study revealed spatial and seasonal variations in the MP pollution status, elucidated potential pollution sources, and identified hotspots linked to visitor use and construction activities for targeted intervention (improve onsite waste collection, visitor surveillance, and the placement of bins and informative signs).

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Introduction

Throughout history, wetlands have offered a wide range of ecosystem services, from nutrient retention to recreation and aesthetics (Boyer & Polasky, 2004). They function as protective zones, areas for the growth and reproduction of wildlife, and to reduce the effects of urban heat islands (Alikhani et al., 2021). However, they are also highly fragile and susceptible to negative impacts such as contamination from hydrocarbons, metals, plastic waste, and microplastics (MPs) (Joshi et al., 2021; Sharley et al., 2017).

Plastics, since their invention, have revolutionized daily life and industrial processes, offering comfort, safety, and hygiene. They are moldable, resistant, durable, inert, lightweight, and low-cost materials. Unfortunately, global demand and production have outpaced the world's capacity for their reduction, recovery, repair, reuse, and recycling. It is estimated that only 9% of plastic waste is adequately recycled (d'Ambrières, 2019); the rest ends up dispersed in various environmental compartments, where it fragments through processes such as weathering and erosion, eventually forming microplastics.

Microplastics (MPs) are defined as plastic micro-particles ranging in size from 1 to 5000 μm . They are classified as primary and secondary. Primary MPs are intentionally manufactured with specific properties to fulfill certain functions and are commonly found in personal care products such as toothpaste, exfoliants, and cosmetics. Some primary MPs are also used as raw materials in manufacturing. In contrast, secondary MPs result from the weathering and degradation of macroplastics, as they are exposed to environmental factors (UV radiation, wind, surface runoff, chemical and biological reactions), which promote their aging, fragmentation, and dispersion (Dawson et al., 2018; Miranda et al., 2021; Stride et al., 2024).

Research on MPs has surged in recent years, driven by their environmental ubiquity, persistence, and potential effects on living organisms. MPs have been found in diverse environments, from Antarctic ice to deep ocean trenches (Kelly et al., 2020;

Van Cauwenberghe et al., 2013; Zhang et al., 2020). Their abundance in different environments varies depending on the type of human activities in the area, seasonality, and environmental factors that influence their mobility and transport (Dai et al., 2018; Dey et al., 2023; Warriar et al., 2022). MPs are often ingested by organisms that mistake them for food, and several studies have demonstrated their transfer along the food chain (Kataria et al., 2024; Saeedi, 2024).

Monitoring microplastic pollution has focused primarily on marine environments, as these are one of the main destinations for land-based runoff. However, microplastics are ubiquitous pollutants, and their presence in non-marine environments, such as rivers, streams, and lagoons is significant. Rivers are recognized as key vectors for transporting microplastics from inland areas to coastal zones and, ultimately, to the ocean. However, they are not merely passive transporters; they also act as deposition sites, with rates varying depending on polymer density and changes in the energy of the river system. Interactions between riparian flora and fauna populations and these contaminants have also been observed (He et al., 2021; O'Connor et al., 2022; Owowenu et al., 2025). Furthermore, lagoon environments have been classified as microplastics sinks. However, factors such as the buoyancy of polymers, their interaction with benthic and planktonic fauna and vertical mixing processes, in which wind, temperature and changes in relevant physicochemical parameters play a significant role, result in complex dynamics within these environments; therefore, classifying them solely as sinks is an oversimplification (Bhutto et al., 2023; Doto et al., 2025; Stride et al., 2024). In all cases, the authors attributed the presence of MPs to human influence. In this regard, urban wetlands are highly vulnerable because they are in constant and close contact with human populations that generate waste. This can enter these systems through the air, runoff, wastewater discharge, and the disposal of plastic waste, among other means (Lei et al., 2023). In this context, the present study is the first to report MPs' presence in the surface waters of the urban wetland "Laguna del Camarón", Sinaloa, Mexico. The main objectives were: 1) to identify possible main sources of this type of pollution in the system through the morphological and chemical characterization of the microplastics found; and 2) to assess the variability in abundance and distribution of microplastics in surface waters during two seasons, considering that, their release rate from potential sources, as well as the

hydrological conditions inside the lagoon change considerably between the dry and rainy seasons.

Materials and methods

Study area and sampling method

The wetland Laguna del Camarón, from now on called LC, is located within the limits of the Central

Park Modernization Project, in the city of Mazatlán, Sinaloa, in northwestern Mexico (Fig. 1), and it has an area of approximately 5.64 ha. LC maintains a permanent underground connection with the sea; it is a highly intervened wetland that still provides various ecosystem services, such as flood regulation, species refuge (local and migratory), and provides landscape, cultural, and tourist benefits. However, like many water bodies worldwide, it has been vulnerable to anthropogenic impacts. Mazatlán

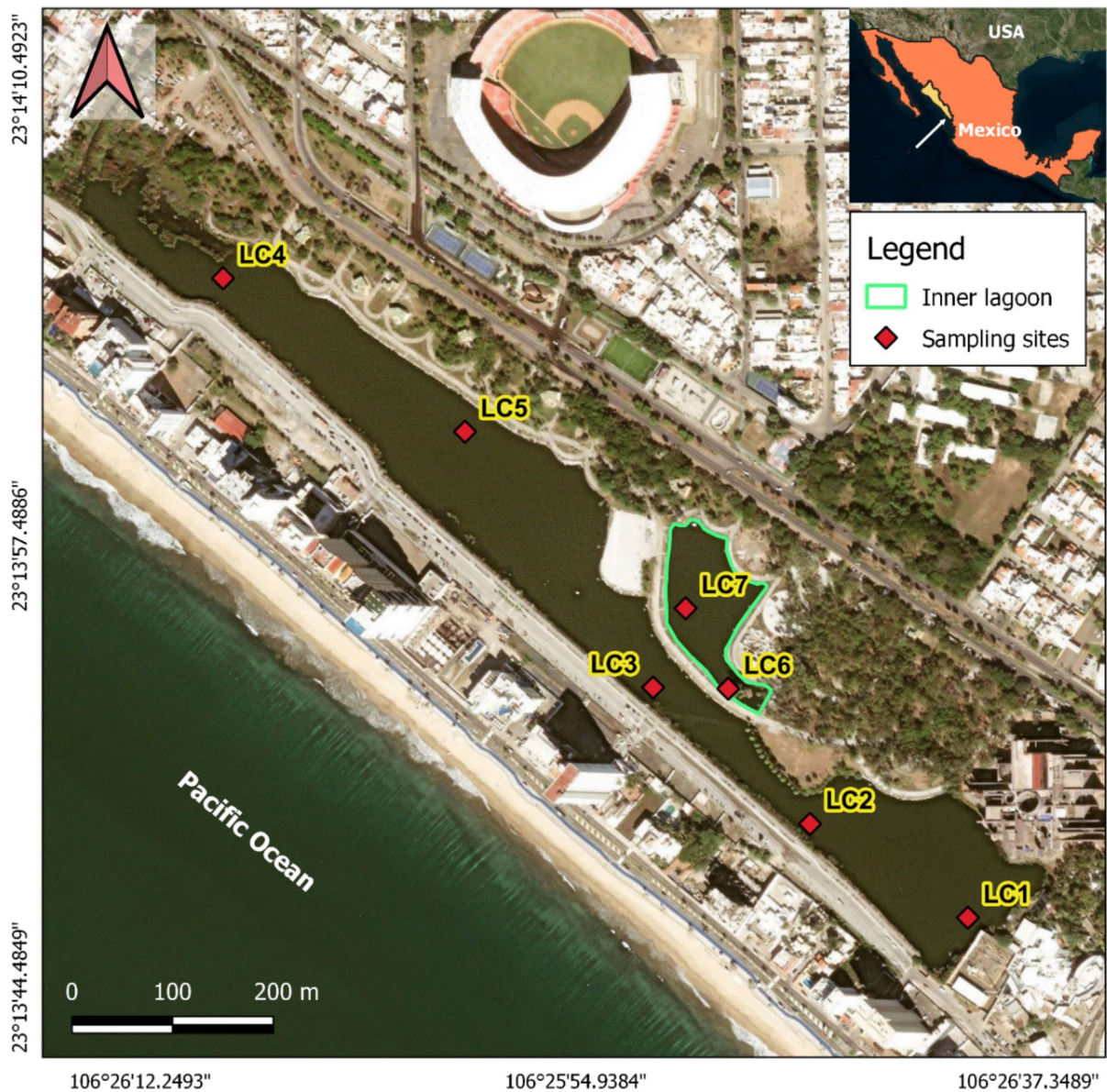


Fig. 1 Study area and sampling sites location in the urban wetland Laguna del Camarón (LC), in northwestern Mexico

is one of Mexico's most prominent tourist destinations in recent years, and the current surge in hotel demand has led to large-scale development along its coastal zone.

The LC system is directly affected by tourism-related activities, including hotels, condominiums, restaurants, ongoing construction, vehicular traffic, and recreational areas. Following the park's modernization, LC system was divided into two distinct water bodies: (a) the main lagoon, and (b) a smaller internal lagoon, situated in the area with the highest visitor concentration.

Seven sampling sites were selected along the LC system—five in the main lagoon and two in the internal lagoon (Fig. 1). Sampling was conducted in October 2022 and March 2023, corresponding to the rainy and dry seasons, respectively. At each site, three replicate samples were collected. All sampling was carried out from a small kayak-type vessel. Approximately, thirty liters of water were collected by replicate at each site, 15–20 cm depth, using a pre-cleaned glass container of 2 L. After pouring the samples through the sieve, the sampling jar was rinsed with filtered water, and sample plus rinse water were filtered through a stainless-steel mesh sieve with a pore size of 63 μm , which was rinsed with filtered water between replicate and sampling site, therefore, only the fraction from 63 to 5000 μm was considered in this work. After filtration, the sieve mesh was rinsed with filtered triple-distilled water, and the rinse water was collected in glass jars. All samples were transported and stored at 4 °C in the laboratory until further analysis.

Sample treatment

In the laboratory, all samples were removed from refrigeration and placed inside a laminar flow hood, where they remained sealed until reaching room temperature. To extract microplastics, a digestion process was carried out, as organic matter can interfere with microplastic recovery. A volume of 100 mL (1:1 v/v) of filtered hydrogen peroxide (H_2O_2 , 30%) was added to each sample. The mixture was stirred at 80 rpm and maintained at 60 °C overnight. In this regard, Masura et al. (2015) and Prata et al. (2019) have shown that 30% peroxide and the application of temperature remove excess organic matter; which facilitates the release of microplastics and does not affect the polymeric structure of the particles.

After digestion, the sample was decanted and filtered through glass fiber filters (Whatman™ GF/F), with a pore size of 0.7 μm , using a vacuum pump (Rocker 300). To ensure total particle extraction and remove any particles adhering to the flask, these were washed with filtered water and then incorporated into the filter. The filters were then placed in glass Petri dishes and dried at 40 °C for 2 h. Once dried, the filters were sealed in the Petri dishes and wrapped in aluminum foil for storage until further observation.

Observation and morphological characterization of microplastics

Each Petri dish was carefully unwrapped under a stereomicroscope to isolate the MPs. The filters were examined horizontally at a maximum magnification of 5X. Potential MPs were isolated onto a new, clean paper filter. Each paper filter was folded in half and wrapped in aluminum foil for storage until chemical analysis. Although handling was minimized, folding the filters may have caused particle loss or transfer, which could have biased the reported results. During the observation phase, the number of potential MPs, along with their morphological characteristics—such as color and shape—were recorded, following Hidalgo-Ruz et al. (2012). MPs retained by the stainless-steel mesh (63 μm) were counted and characterized according to their shape, color, and polymer composition. Microparticle size measurements and classification into size intervals were not included in the scope of the present study.

Chemical characterization of microplastics

The polymer identification of MPs was performed using Fourier-transform infrared (FTIR) spectroscopy (Thermo Scientific™ Nicolet™ iN10 MX), equipped with a μATR accessory with a germanium tip and an MCT detector cooled with liquid nitrogen, capable of measuring particles down to 10 μm . Each spectrum was acquired in the wavenumber range of 600–4000 cm^{-1} , at a resolution of 4 cm^{-1} , with 16 scans per sample. Due to the total number of particles isolated as potential microplastics, as well as analytical and budgetary constraints, it was not feasible to identify the polymer composition of 100% of the particles. Ten percent of the particles, per site, (200

in total) were analyzed by μ FTIR, using a random subsampling method, in Windows ExcelTM. For the μ FTIR analysis, microparticles were picked up using tweezers and a stainless-steel dissecting needle, and each one was placed on a glass slide.

All spectra were visually inspected and analyzed using the OMNICTM Picta software and compared against the following libraries: (1) HR Spectra Polymer and Plasticizers by ATR (Thermo Fisher Scientific Inc. for Nicolet FT-IR), (2) Hummel Polymer Sample Library (Thermo Fisher Scientific Inc. for Nicolet FT-IR), and (3) Microplastics_ICML (in-house library). Spectra with a match rate greater than 70% were accepted, and the polymeric identity of each MP particle was confirmed (García et al., 2021; Schymanski et al., 2021). In this study, the matches obtained were greater than 80%. Polypropylene and polyethylene microparticles (size > 63 μ m) were used as a reference material to estimate the variability coefficient of the μ FTIR analysis (< 1%; $n=6$).

Pollution load index

The pollution load index (PLI) was used to evaluate the pollution status, per site and for the entire wetland, in terms of microplastic occurrence. To calculate the PLI of the zone (Eq. 1), it is necessary first to determine the contamination factor at each site (CF_i ; Eq. 2) and the PLI per site (PLI_i ; Eq. 3); the equations are the following:

$$PLI = \sqrt[n]{PLI1 \times PLI2 \dots PLIn} \quad (1)$$

$$CF_i = \frac{[MPs]_i}{[MPs]_{background}} \quad (2)$$

$$PLI_i = \sqrt{CF_i} \quad (3)$$

where CF_i and $[MPs]_i$ are the contamination factor and the occurrence of microplastics for each site. $[MPs]_{background}$ represents the background value, due to the lack of information in the area, we used the lowest MP concentration observed in this study as the reference background information (Kang et al., 2024; Li et al., 2020; Nguyen et al., 2023; Ranjani et al., 2022). The CF values represent the contamination factor in each site; thus, n is the number of sampling sites.

The PLI classification is as follows: PLI 0 background concentration, 0—1 unpolluted, 1—2 moderately to unpolluted, 2—3 moderately polluted, 3—4 moderately to highly polluted, 4—5 highly polluted, PLI > 5 very highly polluted.

Statistical data analysis

Differences in the mean concentration of MPs among the seven sampling sites (LC1–LC7) and between the two climatic seasons (rainy and dry) were evaluated by means of a two-way factorial ANOVA, considering site, season, and their interaction as fixed factors. Because microplastic concentration data are typically right-skewed and, in this case, displayed heterogeneous variances among cells, the values were square-root transformed ($\sqrt{x}$) prior to analysis to reduce the skewness of the distribution and to stabilize the variance, thereby meeting the assumptions of the parametric model. The homogeneity of variances of the transformed data was subsequently verified with Cochran's C test for each source of variation (site, season, and the site $\times$ season interaction). When the ANOVA detected significant differences, pairwise comparisons of the means were carried out using Tukey's Honestly Significant Difference (HSD) test for equal sample sizes. The level of significance was set at $\alpha=0.05$, and exact probabilities are reported throughout. All analysis were performed in STATISTICA version 12 (StatSoft Inc., 2014).

Quality control

To ensure the quality of the results and prevent sample contamination throughout all steps of the procedure, only glass and stainless steel were used, all materials employed were pre-washed and cleaned with filtered triple-distilled water, wrapped in aluminum foil, and stored until use. In addition, the laboratory staff wore cotton clothing and lab coats. To avoid airborne contamination, the entire workflow was performed in an isolated area and under a flow cabinet. The lab reagents (e.g., water, H_2O_2) were filtered (0.7- μ m pore size WhatmanTM glass fiber filter grade GF/F) to be microplastic-free. All workbenches and surfaces were cleaned with isopropyl alcohol. Additionally, laboratory controls consisting of filters placed in open Petri dishes were distributed throughout the work areas to account for possible contamination of the

workspace. Furthermore, ten laboratory blanks were processed alongside the samples to determine the limit of detection (LOD) of the analysis (Schymanski et al., 2021; Kutralam-Muniasamy et al., 2023). All the blanks were filtered, visually inspected and the microparticles were counted. The number of microparticles in the blanks range from 2 to 8, with an average of 4.2. In our study, the LOD calculated was 6.5 particles.

Results and discussion

Microplastics abundance

A total of 740 MPs particles were found in the surface water of the entire LC during the rainy season, with an average concentration of 1.17 ± 0.56 particles/L. In contrast, significantly higher concentrations were observed during the dry season, averaging 2.05 ± 2.41 particles/L. The minimum, maximum, and mean values of MPs concentrations at each sampling site are shown in Table 1. These values ranged from 0.00 (LC1) to 3.30 (LC6) particles/L during the rainy season and from 0.00 (LC4) to 7.63 (LC7) particles/L during the dry season. The spatial distribution of MPs concentrations in LC, during both climatic seasons, is shown in Fig. 2.

After the square-root transformation, Cochran's C test confirmed that the variances were homogeneous for all sources of variation (site: $C=0.317$, critical value=0.435; season: $C=0.804$, critical

value=0.999; site $\times$ season interaction: $C=0.108$, critical value=0.162; $p>0.05$ in every case), so the assumptions required by the parametric model were satisfied and the two-way ANOVA could be applied to the transformed data.

Results from the two-way ANOVA revealed significant differences in MP concentration among sampling sites ($F 6, 28=11.97$, $p<0.001$), between climatic seasons ($F 1, 28=4.38$, $p=0.046$), and for the site $\times$ season interaction ($F 6, 28=6.44$, $p<0.001$) (Supp. Table A). The Tukey HSD post hoc test showed that this pattern was driven almost entirely by the two sites of the inner lagoon (LC6 and LC7) during the dry season. Considering the site factor (pooled across seasons), LC6 and LC7, which are more isolated but connected to the main lagoon, differed significantly from each of the five sites of the main lagoon (LC1–LC5; $p\leq 0.016$ in all comparisons) but not from each other ($p=0.927$), whereas the five main-lagoon sites were statistically indistinguishable among themselves ($p>0.86$ in all comparisons). For the season factor, the mean concentration was significantly higher in the dry season than in the rainy season ($p=0.046$).

During the dry season, the mean concentration in sites LC6 and LC7 was significantly higher, with the mean concentration in site LC7 being the highest. The highest concentration, recorded at LC7 during the dry season, differed significantly from every other site–season combination ($p\leq 0.009$) except LC6 during the dry season ($p=0.239$), whereas LC6 in the dry season differed from the least-contaminated cells

Table 1 Concentration levels (particles/L) of microplastics in surface water of the Laguna del Camarón, Mazatlán, Sinaloa

Sites	Rainy season				Dry season			
	Max	Min	Average*	SD	Max	Min	Average*	SD
LC1	1.27	0.00	0.54 ^{a,I}	0.65	1.97	0.43	0.99 ^{ab,I}	0.85
LC2	1.30	0.77	0.98 ^{a,I}	0.28	1.37	0.83	1.02 ^{ab,I}	0.30
LC3	0.83	0.50	0.71 ^{a,I}	0.18	1.17	0.53	0.79 ^{a,I}	0.33
LC4	1.37	0.97	1.22 ^{a,I}	0.22	0.73	0.00	0.38 ^{a,I}	0.37
LC5	2.40	0.43	1.21 ^{a,I}	1.05	0.67	0.50	0.61 ^{a,I}	0.10
LC6	3.30	1.17	2.28 ^{a,I}	1.07	5.73	1.73	3.68 ^{bc,I}	2.00
LC7	1.70	1.03	1.28 ^{a,I}	0.37	7.63	6.00	6.90 ^{c,II}	0.83

*Different lowercase letters denote significant differences among sampling sites within the same season (Tukey HSD, $p<0.05$), and different Roman numerals denote significant differences between seasons within the same site. Means that share at least one letter, or the same numeral, are not significantly different. The significance notation was derived from Tukey HSD comparisons performed on the square-root-transformed data, whereas the tabulated values are arithmetic means on the original scale (particles/L). Max, maximum; Min, minimum; SD, standard deviation

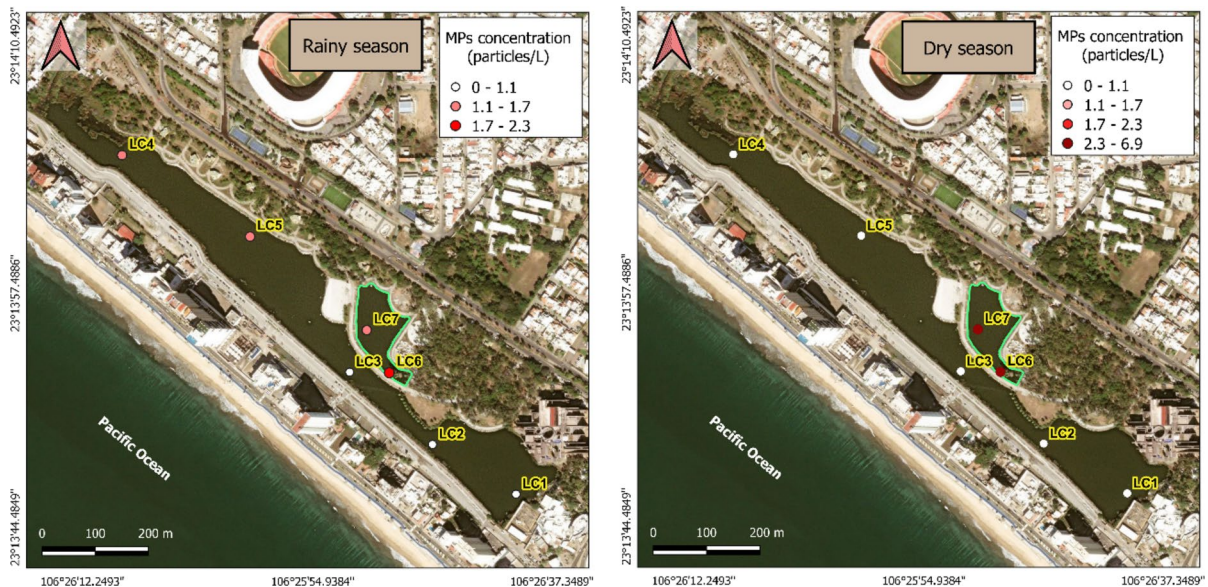


Fig. 2 Microplastic concentrations (particles/L) in water from the urban wetland Laguna del Camarón (LC), during rainy and dry seasons

(LC4-dry, LC1-rainy, LC5-dry, LC3-rainy, and LC3-dry; $p \leq 0.038$). None of the remaining site-season combinations differed significantly from one another ($p < 0.05$). In other words, the rainy-season concentrations at LC6 and LC7 were comparable to those recorded across the rest of the wetland, and it was only during the dry season that the inner lagoon (and particularly LC7) became distinctly enriched in MPs (Table 1). The complete pairwise probabilities for the three sources of variation are provided in Supp. Tables B, C, and D. The observed presence of a large biomass of *Najas marina* in the main lagoon during dry season could be the reason for these differences. Aquatic vegetation can act as a barrier to microplastics, which can reduce their transport across the water body, as indicated by de Los Santos et al. (2021) and Martuscelli et al. (2025) in previous studies. This could explain why there is a relatively higher abundance of microplastics in LC6 and LC7 during the dry season.

It is also important to note that the inner lagoon, where sites LC6 and LC7 are located, is very close to the area with the most significant visitor influence, including dining areas, the main outdoor stage for artistic events, as well as cycling, skating, and athletics paths. This area is a potential source of plastic waste and subsequently MPs. High MP concentrations have been reported, attributing them to

recreational activities, tourism, and other activities with poor disposal and, consequently, accumulation of plastic debris (Bayo et al., 2019; Chandrakanthan et al., 2024). Therefore, one might expect greater airborne particle deposition; however, further studies in this area taking this into account would need to be conducted. Lei et al. (2023) studied MPs contamination in wetlands with different land uses, and they observed a direct relation between MPs concentrations and the distance with urban economic centers. They mentioned that this is mainly due to increased accumulation of plastic waste, vehicle transport routes, urbanization level, population density, and other human activities.

Furthermore, slightly higher concentrations of MPs were observed in the rainy season at sites LC4 and LC5, which are located in the main freshwater inflow zone of the main lagoon. This is due to a slope that causes greater runoff, carrying out debris from nearby streets during meteorological events. At the remaining sites, the distribution of MP concentrations was similar in the rainy and dry seasons. While several studies have demonstrated the influence of surface runoff following meteorological events on the accumulation of plastic debris (Wang et al., 2022; Cho et al., 2023; Shafi et al., 2024), this trend was only observed in LC4 and LC5 in this study. Gerolin

et al. (2024) demonstrated that increasing the water volume controlled by the Guarapiranga reservoir dam reduces the concentration of MPs in surface runoff during the rainy season, generating a dilution effect on the contamination. Further studies are recommended to consider the evolution of MP discharge throughout an entire precipitation season in this study area.

Microplastic pollution in wetlands has been documented worldwide in numerous studies. The concentrations found in LC are higher than those reported for some Colombian wetlands (Table 2). Garcés-Ordóñez et al. (2022) investigated microplastic pollution in surface waters and reported concentrations between 0 and 0.30 particles/L, in a wetland from Colombian Caribbean, they found that the highest microplastic pollution occurs near urban areas and river mouths (Table 2). Another investigation in Colombia reported consistently low concentrations; Porras-Rojas et al. (2023) investigated microplastics in water from Gualí Wetland, and they report concentrations between 0.17–0.31 particles/L, in three sites under the influence of different anthropogenic activities.

By contrast, there are studies reporting results of an order of magnitude higher than those of the present study. Kadir et al. (2025), in Australia, studied water

from the Centennial Parklands pond catchment and found concentrations ranging from 3.1 to 11.9 particles/L (Table 2). Similar results were reported by Zhang et al. (2025) in surface waters from an urban wetland in China; they report concentrations of 2.4–14.9 particles/L.

Forero-López et al. (2021) analyzed MPs in a highly impacted estuarine wetland in Argentina; they collect the water samples using a Nansen net of 60 µm, unlike the other studies mentioned earlier, and found concentrations from 0.182–33.733 particles/L, which is higher than all the compared studies.

Microplastic characterization

Color of microplastics

In LC, a wide range of colors was observed in MPs particles, as shown in Fig. 3. Predominantly, black, brown, blue, white, and yellow were the most common colors. In contrast, other colors such as transparent, green, red, gray, orange, pink, and purple were observed in lower proportions during both the rainy and dry seasons. We also found several particles showing a multicolor pattern, that is, they have two or more colors in their structure.

Table 2 Comparison of the present study with other studies conducted on wetlands around the world

Location	Sampling method	Polymer identification	Abundance (particles/L)	Shape	Polymer	Reference
Bahía Blanca, Argentina	Nansen net of 60 µm, mean speed=0.7 m ⁻¹ for 15 min	FTIR	0.182–33.733	Film, fragment, fiber, and pellets	Rayon, viscose, and textile cotton	Forero-López et al. (2021)
Ciénaga Grande de Santa María, Colombia	100 L filtrated in situ with 300 µm sieve	ATR-FTIR	0–0.30	Fiber, fragment, film, foam, and granule	PP, PE, Nylon-12, and PS	Garcés-Ordóñez et al. (2022)
Gualí Wetland, Colombia	36 L filtrated in situ with 25 µm sieve	ATR-FTIR	0.17–0.31	Fragment, pellet, and foam	HDPE, CPVC, PET, PPS, and others	Porras-Rojas et al. (2023)
Centennial Parklands, Australia	30 L filtrated in situ with 20 µm sieve	µ-FTIR	3.1–11.9	Fragment, film, fibre, and foam	CPE, HDPE, PP, PEMA, PE, PS, and others	Kadir et al. (2025)
Sanyang wetland, China	67.5 L sampled with portable pump	µ-FTIR	2.4–14.9	Fiber, fragment, and particle	PET, PP, and PE	Zhang et al. (2025)
Laguna del Camarón, Mexico	30 L filtrated in situ with 63 µm sieve	µ-FTIR	0–7.63	Fibers, fragments, foam, and film	PES, PP, PE, and PS	This Study

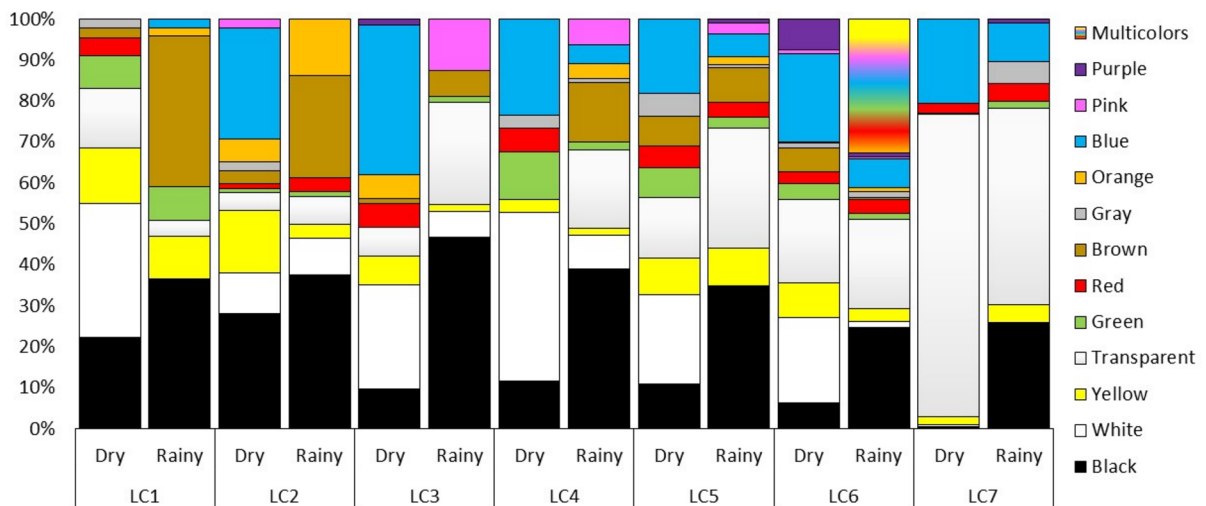


Fig. 3 Range of colors (as percentages) of microplastics found in surface water of “Laguna del Camarón (LC)” during two climatic seasons

Similar results were reported by Ganesan et al. (2019) in their study on MPs rates and transport in different water types, and by Ronda et al. (2023) in their work on the effect of tourism and coastal recreation on MPs pollution in beaches of the southern Buenos Aires province. The authors found that white, green, yellow, brown, blue, and black were the predominant colors. The predominance of black and blue microfibers may be related to the deposition of textile micro-particles from the clothing of visitors frequenting LC. Turner et al. (2019) mentioned that the presence of black and blue microparticles corresponds to the indigo dye used in textile staining, however, this must be corroborated using other chemical techniques that allow for the identification of textile dyes.

In general, a higher relative abundance of black particles was observed at all sites during the rainy season respect to the dry season. This is the only consistent pattern across all sampling sites in relation to the colours detected. Therefore, it can be assumed that the influx of black particles is linked to runoff. Several authors have indicated that the color of MPs is a parameter influencing their accumulation in marine organisms, as dark and bright-colored MPs can be mistaken for food (Alvarado-Zambrano et al., 2023; Manikanda et al., 2021).

Today, plastic debris of various colors can be found in the environment, however, color alone does not provide much information about the origin of the microparticles; it is necessary to find a way

to integrate color and chemical composition data in order to identify pollution sources more accurately.

Shape of microplastics

Fibers, foam, fragments, and film were the shapes observed among the MPs found in LC during both climatic seasons. Fibers accounted for more than 60%, followed by foam with approximately 20%, fragments with around 15%, and film with less than 5% (Fig. 4). The predominance of fibers, as previously mentioned, can be associated with the contribution of microfibers originating from visitors' clothing (Ronda et al., 2023), which can be easily transported by wind (Xiao et al., 2023).

At sites LC6 and LC7, the presence of microfibers exceeded 75%, which may reflect the high influx of visitors to the study area. Meanwhile, the occurrence of foam and fragments can be associated with 1) the accumulation of single-use plastic waste derived from food packaging and recreational activities and 2) the intense construction activity of hotels, condominium buildings, and family recreational facilities along the reservoir's margins. In the present study, a tendency of increase in microparticles with foam and fragments shapes in the main lagoon (from LC5 to LC1) was observed (Fig. 4), and further studies should be conducted to describe this behavior. It is important to mention that construction activities were taking place

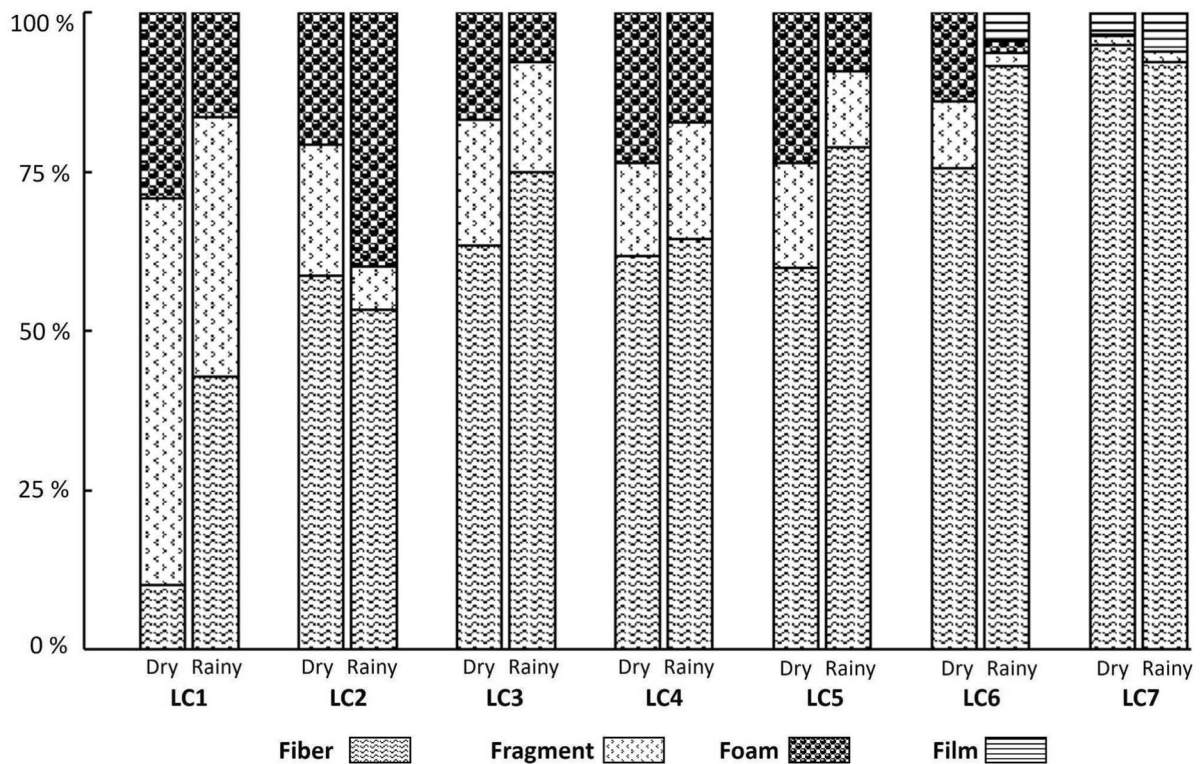


Fig. 4 Distribution of microplastic shapes (%) in the surface water of Laguna del Camarón (LC) during two climatic seasons

in the vicinity of LC1. Foam, in its various forms, is a material widely used in single-use containers, food packaging, and as an insulation material in building and construction due to its physical and chemical properties (Gao et al., 2023; Turner, 2020).

Polymeric composition

Overall, in both seasons, polymer identification showed the presence of polyester (PES, 48% and 62%), polypropylene (PP, 19% and 23%), polyethylene (PE, 10% and 2%), and polystyrene (PS, 5% and 5%), which were the most dominant polymer types (Fig. 5) in dry and rainy seasons, respectively. Other polymers such as polyacrylonitrile (PAN), polyvinyl chloride (PVC), phenolic resin (PR), alkyd resin (AR), acrylic, and polydiallyl phthalate (DAP) were also detected in low proportions ($\leq 5\%$). Natural polymers were also identified in surface water, notably cotton (32 particles across both seasons) and cellulose (29 particles across both seasons); these counts

were not included in the frequency calculations or the comparison with other investigations.

Similar results were observed by Suteja et al. (2021), in their study on the temporal and spatial distribution of MPs in the waters of Benoa Bay, Indonesia, and by Rios-Mendoza et al. (2021) in surface water from Mazatlán Bay, Mexico, both of which highlighted PE, PP, and PS as the dominant polymers.

LC is a recreational and leisure area surrounded by new real estate developments; therefore, the origin of the identified polymers may stem from the accumulation and improper management of waste generated by activities carried out within the park, in addition to shedding synthetic fibers from polyester and cotton from the sports and everyday clothing worn by visitors.

The textile industry primarily uses synthetic fibers of PES, or polyethylene terephthalate (PET), and natural polymers such as cotton in garment production. De Falco et al. (2020) report that direct release

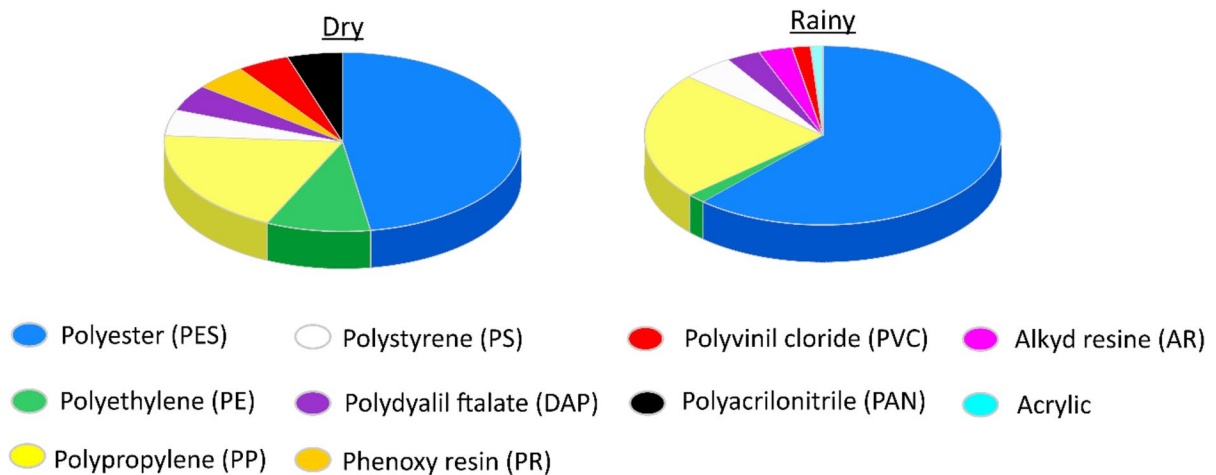


Fig. 5 Proportion of polymers identified in particles from the surface water of Laguna del Camarón (LC) during two climatic seasons

of PES and cotton microfibers from garments to air, from garment wear, is as important as release to the environment from wastewater, one of the most recognized sources of MP pollution.

PE is the most used plastic polymer (Qiang et al., 2023), accounting for around 30% of the total plastic polymers' demand (Plastics Europe, 2018). PE is used in the manufacture of carry bags, bottles, bottle caps, and toys; once in the environment, the wind, rain, and solar UV breakdown the exposed debris to microplastics (Qiang et al., 2023). Meanwhile, PP and PS are polymers commonly used in the production of utensils such as spoons and plates, product packaging, food storage, and construction materials.

Pollution load index

The pollution load index calculated for each site varied from 1.12 to 2.30 in the rainy season, and from 0.94 to 4.01 in the dry season. During both seasons, sites LC6 and LC7 showed the highest values. In rainy seasons LC6 and LC7 showed values of 2.30 and 1.73, classifying them as moderately polluted and unpolluted to moderately polluted, respectively; in the dry season LC6 and LC7 had 2.92 and 4.01 values, classifying them as moderately polluted and highly polluted, respectively. Sites LC6 and LC7 were initially identified as potential hotspots for microplastic pollution, which was later confirmed

by statistical analysis, as they are located in the area most exposed to recreational activities, as described in Section "Microplastics abundance". In terms of the entire zone, for both seasons, LC can be classified as unpolluted to moderately polluted.

These results should be considered by local authorities or managers at these sites to strengthen regulations and increase visitor surveillance.

Limitations

It should be noted that this study includes only surface water measurements; sediments and organisms were not sampled, and quantitative source apportionment was not performed. The polymeric analysis was performed in 10% of the particles. Therefore, our results should be interpreted as a baseline assessment of microplastic contamination and the potential sources of this pollutant in this ecosystem.

Conclusions

The spatial and temporal distribution of MPs in the surface waters of Laguna del Camarón was evaluated. The input of microplastics to this water body may be a result of different processes, such as the aerial deposition during the dry season due to the proximity of LC to roads, hotels, and residential

areas. In terms of seasonality, mean surface-water MPs concentrations for the entire urban wetland were significantly higher in the dry season (2.05 particles/L) than in rainy season (1.17 particles/L); this seasonal difference may reflect higher recreational use and reduced dilution effect during the dry season. The highest concentrations of MPs were observed in the inner lagoon during both climatic seasons, identifying it as a hotspot of MPs pollution. This distribution is likely influenced by the proximity of the wetland to waste accumulation areas resulting from recreational activities within the Central Park and the plant barrier (*Najas marina*), that isolates it from the main lagoon. The shape of the MPs indicated the presence of microfibers (> 60%), likely originating from visitors' clothing, as well as fragments and foam, which may be linked to nearby construction activities. The predominant polymers identified were PES, PP, PE, and PS—materials widely used in textiles, bottle caps, food containers, packaging, and storage. Taking together, the spatial patterns and chemical analysis of the polymers suggest that visitor activities and textile derived microfibers are major contributors to the pollution status of the wetland.

This study constitutes the first assessment of MPs in this area and provides comparable data to studies conducted in similar environments. The findings underscore the urgent need for implementing effective waste management (improve onsite waste collection and the placement of bins and informative signs) and control strategies across all human activities to reduce plastic pollution in aquatic ecosystems, which contribute to the formation of microplastics.

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Author contributions JRRH and DAZ worked in the conceptualization, methodology, and investigation. JRRH worked in the validation, data curation, and writing the original draft. Formal analyses were carried out by JRRH, FAM, and DAZ. JRRH, JHA, and DAZ produced the figures (visualization). The resources were provided by CRGR. All authors contributed to manuscript revisions (Writing—review & editing) and approved the final version.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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